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ECE 410/510: Hardware for AI and ML, Spring 2026

Codefest #7: Sparse MVM + OpenLane output interpretation

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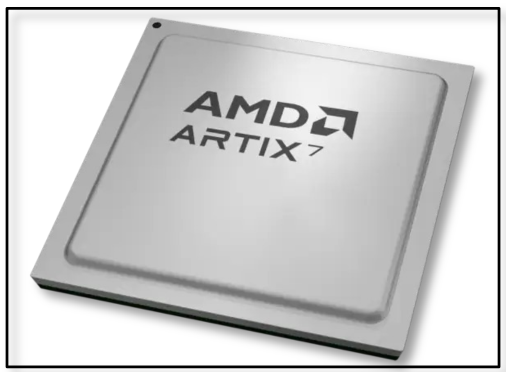

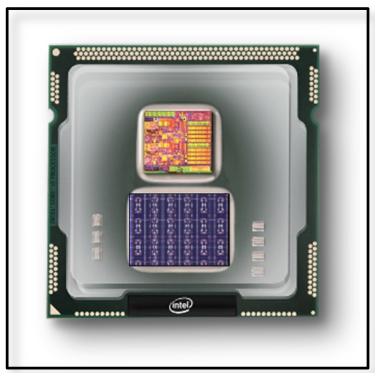


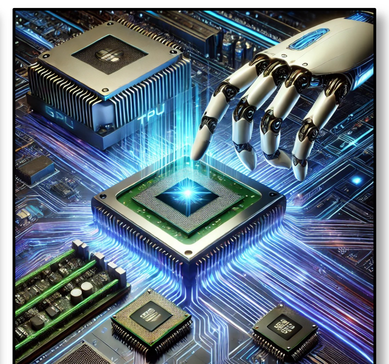
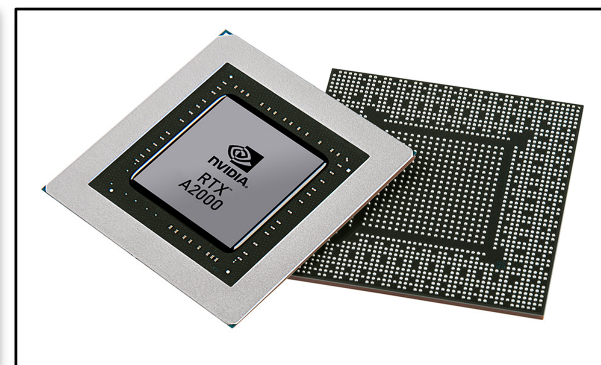

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CSR: Compressed Sparse Row

QUIZ

Example matrix A

	j=0	j=1	j=2
i=0	5	0	0
i=1	0	0	8
i=2	0	3	0

`values` *the nonzeros*

5	8	3

`col_idx` *column of each NZ*

0	2	1

`row_ptr` *where row i starts*

0	1	2	3

3 is sentinel. It equals the total nonzero count.

- col_idx stores coordinates directly. row_ptr does not.
- col_idx[k] = the column of the k-th nonzero. Direct coordinate, one entry per NZ.
- row_ptr does not store "the row of each NZ." It would be wasteful to repeat the row index for every NZ in the same row. Instead, it stores where each row's run of nonzeros starts in the values array. Length N+1, not nnz.

How to read row i

Slice **`row_ptr[i]..row_ptr[i+1]`** gives the nonzeros in row i.

Row 1: **`row_ptr[1..2] = [1,2]`**

→ one NZ at index 1: **`values[1]=8`** at column **`col_idx[1]=2`**.

Memory: ~2·nnz + N values instead of N^2. Walk rows in order → ideal for y = A·x.

Compared to Coordinate format (COO)

COO stores **`(row, col, val)`** for every NZ — 3·nnz entries.

CSR keeps **`col_idx`** explicitly but compresses the row coordinate into **`row_ptr`** (N+1 entries, not nnz).

= COO with the row column run-length-encoded.

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CSR Example: 4×4 with 2 NZ per row

QUIZ

Matrix A
4×4, 2 NZ per row

	j=0	*j=1*	*j=2*	*j=3*
i=0	5	0	0	3
i=1	0	8	2	0
i=2	6	0	0	9
i=3	0	1	0	7

`values` *the 8 nonzeros, row by row*

k=0	*k=1*	*k=2*	*k=3*	*k=4*	*k=5*	*k=6*	*k=7*
5	3	8	2	6	9	1	7

`col_idx` *column of each NZ*

0	3	1	2	0	3	1	3

`row_ptr` *start of each row in values (N+1 = 5)*

0	2	4	6	8

sentinel = number nz

- **row_ptr[i]** answers "*what value of k does row i start at?*"
- It's a bookmark into the k-indexed arrays.

How to read row i

Slice **`row_ptr[i]..row_ptr[i+1]`** gives that row's nonzeros.

Row 2: **`row_ptr[2..3] = [4,6]`** → NZs at k = 4, 5.

`values[4]=6` at **`col_idx[4]=0`**; **`values[5]=9`** at **`col_idx[5]=3`**.

Storage: 8 + 8 + 5 = 21 ints, vs 16 for dense. CSR wins as N grows.

Read-out: A in (i, j, value) form

```
(0,0,5)  (0,3,3)   row 0 starts at k=0
(1,1,8)  (1,2,2)   row 1 starts at k=2
(2,0,6)  (2,3,9)   row 2 starts at k=4
(3,1,1)  (3,3,7)   row 3 starts at k=6
```

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Reconstructing A from CSR

Matrix A
4×4, 2 NZ per row

	j=0	*j=1*	*j=2*	*j=3*
i=0	5	0	0	3
i=1	0	8	2	0
i=2	6	0	0	9
i=3	0	1	0	7

`values` *the 8 nonzeros, row by row*

k=0	*k=1*	*k=2*	*k=3*	*k=4*	*k=5*	*k=6*	*k=7*
5	3	8	2	6	9	1	7

`col_idx` *column of each NZ*

0	3	1	2	0	3	1	3

`row_ptr` *start of each row in values (N+1 = 5)*

0	2	4	6	8

sentinel = nnz

- **row_ptr[i]** answers "*what value of k does row i start at?*"
- It's a bookmark into the k-indexed arrays.

QUIZ

Walk by row_ptr

```
for i in 0..N-1:
    for k in row_ptr[i]..row_ptr[i+1]-1:
        A[i][col_idx[k]] = values[k]
```

Trace on the 4×4 example:

i=0: row_ptr[0..1]=[0,2] → k=0,1 → **A[0][0]=5, A[0][3]=3**
i=1: row_ptr[1..2]=[2,4] → k=2,3 → **A[1][1]=8, A[1][2]=2**
i=2: row_ptr[2..3]=[4,6] → k=4,5 → **A[2][0]=6, A[2][3]=9**
i=3: row_ptr[3..4]=[6,8] → k=6,7 → **A[3][1]=1, A[3][3]=7**

All 8 NZs placed; A fully reconstructed.

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ASIC design flow

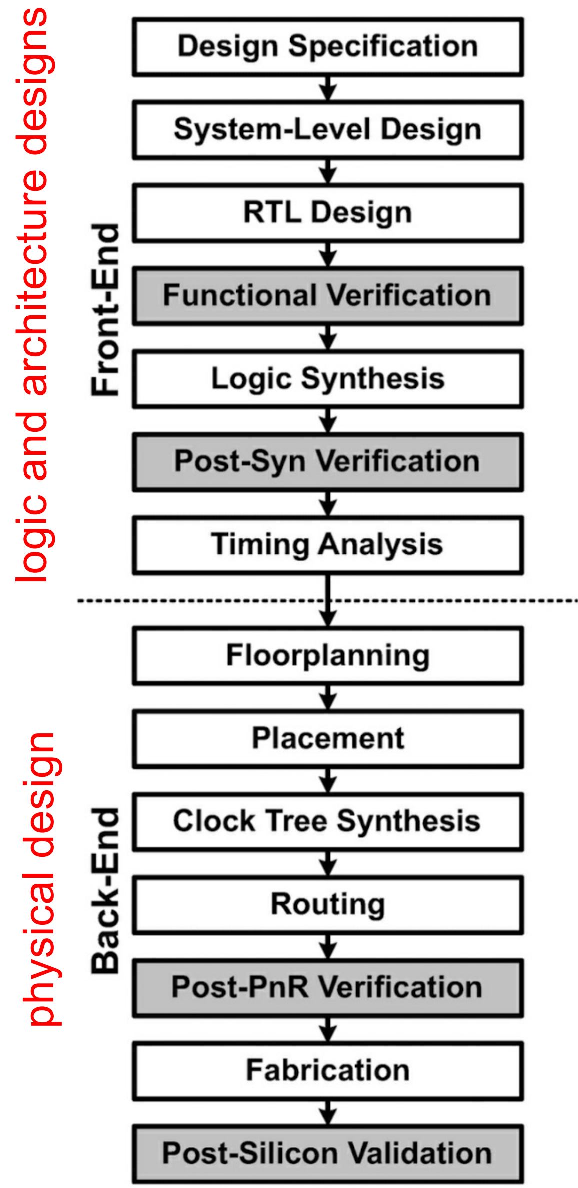

Fig. 9 ASIC design flow

Python
Verilog/SystemVerilog/VHDL

Max. frequency, # transistors, etc.

Mishra, Ashutosh; Cha, Jaekwang; Park, Hyunbin; Kim, Shiho. Artificial Intelligence and Hardware Accelerators.

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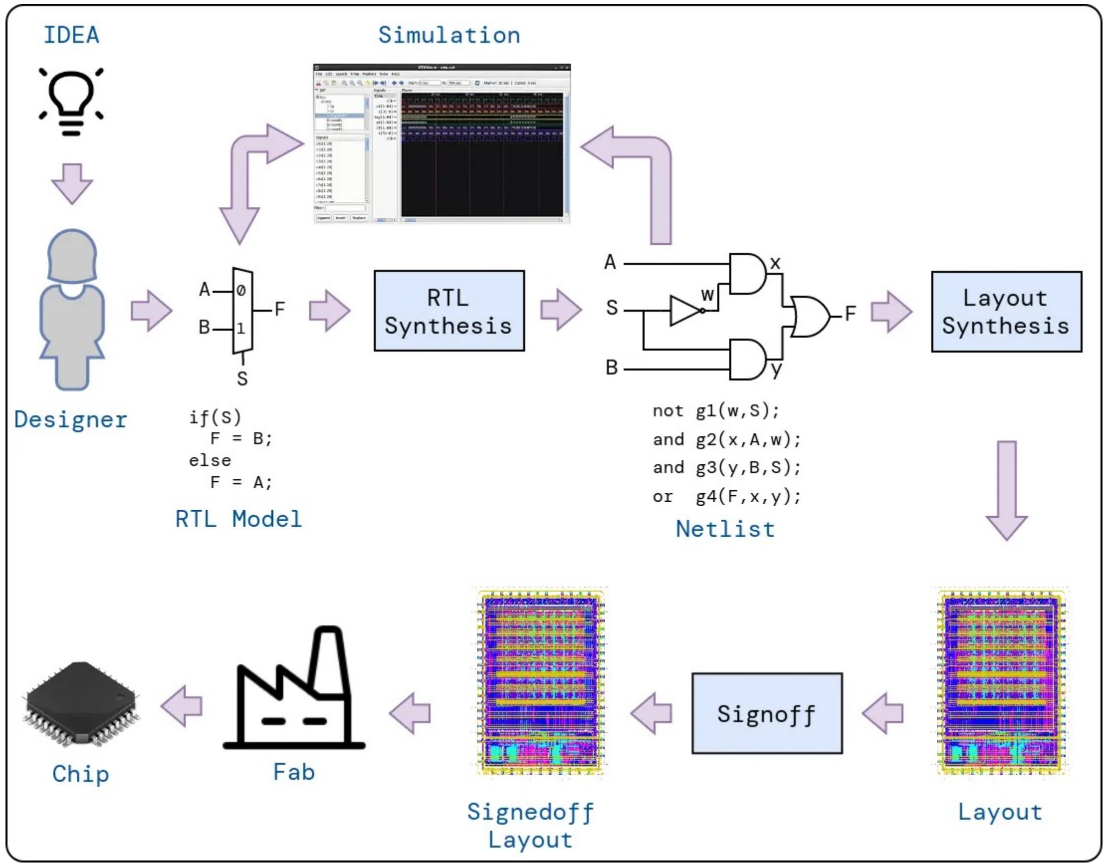
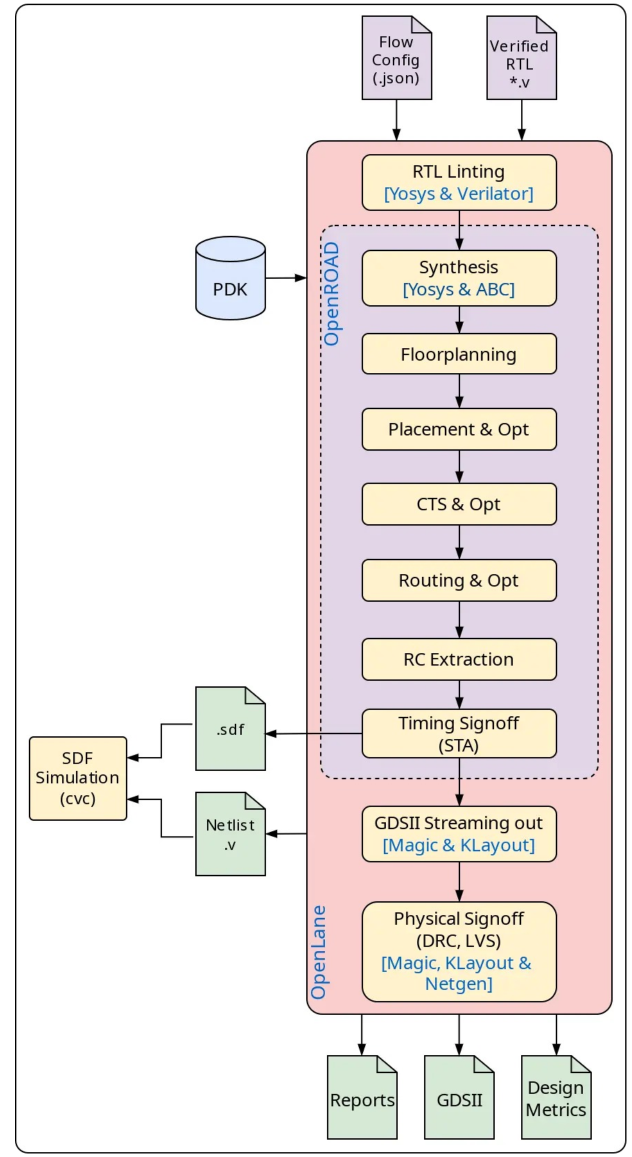
OpenLane 2 tools





https://efabless.com/openlane
https://github.com/efabless/openlane2
https://openlane2.readthedocs.io/en/latest/getting_started/newcomers/index.html

Today's presentations

1 slide, 1 minute. Strict!

Wed, May 13	7	4:55: PM	Stanton, Andrew J. (Pref: Andrew, Pronoun: he/him)
Wed, May 13	7	4:57: PM	Bandla, Karthik
Wed, May 13	7	4:59: PM	Alrandi, Dawood A.
Wed, May 13	7	5:01: PM	Premarajan, Roshan B. (Pref: Roshan, Pronoun: hr/him)
Wed, May 13	7	5:03: PM	Vedantam, Venkata Krishna Kumar (Pronoun: His/Him)
Wed, May 13	7	5:05: PM	Esmaeil, Saleh A. (Pref: Sal, Pronoun: He/Him)